Parasitology in Microbiology: Beginner's Course

Lesson 5: Diagnostic Techniques

Diagnostic Techniques – Study Notes

Key Concepts

1. **Microscopy** – Direct visualization of parasites using light, fluorescence, or electron microscopes.

2. **Staining Methods** – Chemical dyes (e.g., Giemsa, Ziehl Neelsen, trichrome) that highlight parasite structures and improve detection.

3. **Antigen Tests** – Rapid immunoassays that detect parasite derived proteins in clinical specimens.

4. **Polymerase Chain Reaction (PCR)** – Molecular amplification of parasite DNA/RNA for highly sensitive and specific identification.

5. **Sample Collection & Preparation** – Proper specimen type, timing, and handling are critical for all techniques.

Summary

Diagnostic parasitology relies on a hierarchy of methods, from simple visual inspection to sophisticated molecular assays. **Microscopy** remains the cornerstone for many helminths and protozoa; it provides immediate information on morphology, life stage, and parasite load. However, the sensitivity of plain wet mount examination can be limited, so **staining** techniques are employed to accentuate cellular details, differentiate species, and detect intracellular organisms.

When rapid, point of care results are needed, **antigen detection tests** (e.g., lateral flow immunochromatography for *Giardia* or *Cryptosporidium*) offer a quick answer, though they may miss low level infections. For definitive confirmation, especially in low prevalence settings or for cryptic species, **PCR** amplifies target nucleic acid sequences, delivering high sensitivity and the ability to genotype parasites. Understanding the strengths, limitations, and appropriate contexts for each method enables accurate diagnosis and effective disease control.

Important Points

• **Microscopy**

  - Light microscopy (bright field) is sufficient for most stool ova and parasite (O&P) exams.

  - Fluorescence microscopy can detect parasites labeled with specific fluorochromes (e.g., *Cryptosporidium* oocysts with FITC conjugated antibodies).

  - Electron microscopy is reserved for ultrastructural studies, not routine diagnosis.

• **Staining**

  - **Giemsa** – stains nuclei and cytoplasm; ideal for *Plasmodium* spp., *Leishmania* amastigotes, and blood smears.

  - **Trichrome** – highlights intestinal protozoa (e.g., *Entamoeba histolytica*, *Giardia*).

  - **Ziehl Neelsen / Auramine Rhodamine** – acid fast stains for *Cryptosporidium* oocysts.

- Proper fixation (e.g., methanol) preserves morphology and prevents artifact formation.
    • **Antigen Tests**
  - Usually lateral flow or ELISA formats; detect species specific proteins (e.g., Giardia cyst wall protein 1).
  - Provide results in 10–30 /min; useful in field settings.
  - Sensitivity can be 70–95 /%; specificity often >90 /%, but cross reactivity may occur.
    • **PCR Basics**
  - Requires extraction of high quality DNA/RNA from stool, blood, or tissue.
  - Target genes: 18S rRNA (protozoa), ITS regions (helminths), mitochondrial COI (species discrimination).
  - Real time quantitative PCR (qPCR) adds quantification and reduces contamination risk.
  - Controls: positive control, negative extraction control, and internal amplification control are essential.
    • **General Tips**
  - Collect multiple specimens (e.g., three stool samples on separate days) to increase detection probability.
  - Store specimens at 4 /°C for short term, –20 /°C for long term DNA preservation.
  - Combine methods (e.g., microscopy + PCR) for the highest diagnostic yield.

Examples

Example 1 – Detecting *Giardia lamblia* in a stool sample

1. **Microscopy**: Perform a wet mount; look for motile trophozoites. Sensitivity "H 50 /%.
2. **Staining**: Apply trichrome stain; trophozoites appear with characteristic “face like” nuclei and ventral adhesive disc. Improves detection to "H 70 /%.
3. **Antigen Test**: Use a rapid immunochromatographic assay; a positive line indicates Giardia antigen, raising sensitivity to >90 /%.
4. **PCR**: Extract DNA and amplify the 18S rRNA gene; confirms species and can detect sub clinical infections missed by other methods.

Example 2 – Confirming *Plasmodium falciparum* malaria

1. **Blood Smear**: Thick smear for parasite density; thin smear for species identification. Giemsa stain shows ring forms, banana shaped gametocytes.
2. **Rapid Diagnostic Test (RDT)**: Detects HRP 2 antigen; provides result in 15 /min, useful where microscopy is unavailable.
3. **PCR**: Targets the *pfcrt* gene; distinguishes *P. falciparum* from other *Plasmodium* spp. and detects low level parasitemia, essential for surveillance and drug resistance monitoring.

Quick Review

1. **What is the main advantage of using a trichrome stain over a wet mount for intestinal protozoa?**
2. **Name two situations where an antigen test is preferred over microscopy.**
3. **List three essential controls required in a PCR assay for parasite detection.**
4. **Why is it recommended to examine multiple stool samples when diagnosing helminth infections?**
5. **Which microscopy technique is most commonly used for routine O&P exams, and why?**

Further Reading

• **Advanced Microscopy** – Fluorescence and confocal microscopy for intracellular parasites.

• **Molecular Diagnostics** – Loop mediated isothermal amplification (LAMP) and CRISPR based detection methods.

• **Quality Assurance** – Standard operating procedures, proficiency testing, and WHO guidelines for parasitological diagnosis.

• **Epidemiology & Surveillance** – Role of diagnostic techniques in monitoring drug resistance and outbreak investigations.